

W. D. Gann and GSPS

Methodology, Foundation Architecture, and Code-Implementation Audit

Prepared for Charles Coleman | September 8, 2026 | Research perspective: market-methodology analyst and trading-systems architect

Bottom line

GSPS is explicitly conceived around a Gann-inspired numeric layer: decimal stripping, digit-sum reduction modulo 9, and classification around 3 / 6 / 9 “harmonic vectors.” That is a real implementation of a modern Gann-inspired numerological convention, not a faithful implementation of the full W. D. Gann body of work. The provided protocol contains no executable Square of Nine spiral, price–time angle, swing-chart, cycle, volume, or risk-management engine beyond output labels. As a result, Gann is presently a naming and output-framework foundation for GSPS, while most of the historically documented Gann toolkit remains specified conceptually or absent from the code that was available to audit.

Audit scope and limit

This review used: (1) the GSPS/Gann Protocol V6 project instructions supplied with this request; (2) a search of accessible GitHub code for Gann-related implementation; and (3) public primary or near-primary material concerning W. D. Gann, especially his 1923 book *Truth of the Stock Tape* and the 1909 *Ticker and Investment Digest* profile. No identifiable private GSPS repository, branch, source tree, or source files were supplied or discovered through the connected GitHub search. Therefore, the “implemented in code” conclusion is limited to the protocol text and the absence of auditable repository evidence—not a claim that no private code exists.

To convert this into a definitive source-level audit, provide the exact GitHub owner/repository and branch, or a read-only code export. The audit should then search functions, tests, database schemas, prompts, and UI labels for each methodology component listed below.

Who W. D. Gann was

William Delbert Gann (1878–1955) was an American market writer, trader, and course seller associated with early twentieth-century stock and commodity speculation. His enduring reputation rests on a hybrid practice: close reading of price movement and volume; structured trading and money-management rules; geometric and arithmetic price/time studies; and, in later interpretations of his work, calendrical and astronomical cycle studies.

An important distinction is necessary. Gann’s 1909 profile reported exceptionally strong observed trading results—286 transactions during October 1909, 264 described as profitable—but it was a contemporaneous magazine account, not an independently audited, reproducible performance record. It is historically important evidence of his reputation, not proof that a modern system based on any subset of Gann techniques will be profitable. [web:26]

The method in layers

Layer	What Gann emphasized	Practical interpretation for a system
Market structure	Trend, swing highs/lows, resistance, accumulation/distribution, and the relationship between price movement and volume. In <i>Truth of the Stock Tape</i> , he describes tape reading as judging strong versus weak positions and the psychological moment to buy or sell. [web:21]	Use systematic swing detection, trend state, breakout/reclaim logic, relative volume, and market-regime filters.

Layer	What Gann emphasized	Practical interpretation for a system
Price study	Horizontal supports/resistances, prior extremes, percentage/range relationships, and “squares.” Later Gann practice is commonly associated with Square of Nine level projections.	Project price levels from a declared anchor, then record whether price reacts, breaks, retests, or ignores each level.
Time study	Elapsed time since significant pivots and recurring cycle windows. Gann treated time as coequal with price; his book explicitly says the length of time and total point movement both matter in judging accumulation and distribution. [web:21]	Calculate trading-session and calendar elapsed time from objectively selected pivots; make time windows alerts, not automatic entries.
Price–time geometry	Gann angles and “squaring” time with price. The 1x1 concept is only meaningful after defining chart scale or a normalized unit of price per bar.	Use normalized slope: ATR per bar, percentage per bar, or log-price per bar. Never infer a 45-degree market law from screen pixels.
Execution and risk	Waiting for confirmation, stops, scaling/pyramiding carefully, and position discipline. Gann wrote that unclear indications require waiting until the balance of supply and demand is evident, and warned against overlooking volume. [web:21]	Define entry confirmation, fixed invalidation, position-risk budget, partial exits, slippage assumptions, and a no-trade state.
Esoteric / astronomical claims	Commonly attributed to Gann but difficult to separate from later interpretations and marketing. These claims have no established causal or broadly accepted empirical basis.	Treat as optional research hypotheses only; isolate them behind experimental flags and demand preregistered out-of-sample evidence.

What is historically well supported

- Price, time, range, trend, support/resistance, and volume were central to Gann’s practical market writing.
- He advocated record keeping, charts, patience when evidence was unclear, and waiting for resistance/confirmation before taking a position. [web:21]
- The documented 1923 material is materially more operational and testable than the mythology often attached to “Gann theory.”

What needs caution

- A number pattern, a chart angle, or a cycle window is not an entry signal by itself. Gann’s own practical writing repeatedly pairs levels with market action, volume, and confirmation. [web:21]
- Claims that fixed 3, 6, 9 digit roots have predictive force across markets are not established by the historical sources reviewed. A decimal representation changes with quote precision, stock splits, currency denomination, and price scale; that creates an avoidable measurement problem.
- Technical-analysis profitability cannot be assumed from an appealing historical narrative. Any GSPS feature must be tested against transaction costs, survivorship bias, look-ahead bias, data-snooping, and regime dependence.

GSPS: current Gann foundation

The project's stated engine is:

```
clean_str = "".join(c for c in str(current_price) if c.isdigit())
base_sum = sum(int(digit) for digit in clean_str)
current_root = base_sum % 9
if current_root == 0: current_root = 9
```

This function computes a digital root from the digits of the displayed price. For example, a price displayed as 123.45 becomes digits 1+2+3+4+5 = 15 and then root 6. It is deterministic and easy to reproduce, but it is not the mathematical mechanism of the traditional Square of Nine, which uses a spiral-number geometry and angular transformations from an anchor. It also does not compute a Gann angle, time cycle, or price–time square.

The protocol then asks for “nearest 3, 6, and 9 Harmonic Vectors,” material-number/harmonic-node classification, and precise entry, take-profit, master-target, and stop-loss coordinates. Those outputs are currently under-specified: no formulas establish how price roots translate into a distance, node, target, or hard stop. Without those definitions, the system risks producing precise-looking prices that cannot be reproduced or validated.

Implementation coverage assessment

Gann component	Evidence in GSPS protocol / code audit	Status	Coverage estimate
Digit reduction / mod 9	Explicit function provided: decimal stripping, digit sum, modulo 9, map zero to 9.	Implemented in specification	100% of this narrow feature
3 / 6 / 9 node taxonomy	Protocol explicitly calls these harmonic vectors/nodes, but offers no distance metric, scoring logic, or level-generation formula.	Defined conceptually; not computationally complete	About 25%
Square of Nine	No spiral index, anchor, square-root rotation, or 45/90/180/270-degree projection formula identified.	Not evidenced	0%
Gann angles / fan	No anchor-pivot selection, price-per-bar normalization, slope calculation, intersection logic, or chart scale specification identified.	Not evidenced	0%
Time cycles / anniversaries	No elapsed-bar/session calendar engine or cycle confluence logic identified.	Not evidenced	0%
Swing chart / pattern rules	No fixed-box or percentage reversal rules, swing sequence, higher-high/lower-low state machine, or pivot confirmation algorithm identified.	Not evidenced	0%
Volume / supply-demand confirmation	Although core to Gann's practical writing, no volume condition appears in the supplied protocol.	Not evidenced	0%
Entries, targets, stops	Required output fields exist, but no calculation or risk parameter creates their coordinates.	Output schema only	About 10%
Validation / backtesting	No data protocol, benchmark, split, costs, performance metric, or rejection rule provided.	Not evidenced	0%

Composite conclusion: based strictly on the material available, GSPS currently implements roughly 10–15% of a practical, testable Gann-inspired research stack. That percentage is qualitative—not a code-line count—and it refers to methodological coverage. It is 100% complete only for its narrow decimal-strip modulo-9 routine. It is not yet a full Gann system.

Why this matters

The current GPS protocol gives Gann language a high decision-making role while giving the underlying computations little definition. The remedy is not to remove the Gann layer. The remedy is to turn every claim into a transparent feature, every feature into a testable hypothesis, and every trading output into a traceable decision record. That preserves the conceptual foundation while avoiding unverifiable signal generation.

Recommended GSPS architecture

GSPS should be built as a layered decision-support system—not an oracle. The Gann module should create candidate levels and time windows; independently measured market structure and risk controls should decide whether a trade is eligible.

Module	Inputs	Outputs	Required test
1. Data normalization	OHLCV bars, corporate-action-adjusted prices, session calendar, symbol metadata.	Canonical price scale, bar index, volatility measures.	Reproducibility across split-adjusted and unadjusted data; no formatting dependence.
2. Pivot and trend engine	Normalized bars, lookback/reversal threshold.	Confirmed pivots; trend regime; swing range; anchor confidence.	No look-ahead pivot confirmation; evaluate delay and false-pivot rate.
3. Gann price engine	Pivot anchor, price scale, optional Square of Nine rotations, prior ranges.	Named projected levels with source and formula.	Reaction frequency versus matched non-Gann control levels.
4. Gann time engine	Pivot dates/bars, candidate intervals, market sessions.	Time windows and confluence score.	Forward return/reversal distribution inside vs. outside windows.
5. Geometry engine	Anchor, ATR or percent/log-price normalization, bars elapsed.	1x1 and related normalized trend lines, cross states.	Out-of-sample predictive value over simple trend baselines.
6. Confirmation engine	Trend state, level proximity, time confluence, relative volume, volatility, breadth.	Long/short/no-trade eligibility and reason codes.	Ablation tests: measure incremental value of each feature.
7. Risk engine	Entry, invalidation level, ATR, account-risk parameter, liquidity/cost assumptions.	Position size, stop, targets, expected R multiple, expiry.	Maximum loss and drawdown constraints under stress scenarios.
8. Audit and research ledger	All inputs, parameter versions, signal decisions, realized outcome.	Immutable signal card and performance report.	Walk-forward and paper-trading reconciliation.

A usable signal contract

Every GSPS signal should state: symbol; timeframe; data timestamp; selected pivot and its confirmation timestamp; active trend; each projected Gann level with formula; time-window status; volume/market-structure confirmation; exact entry trigger; stop/invalidation; targets; risk per share; position sizing assumptions; model version; and an explicit no-trade reason when confluence fails. This makes a “Gann Coordinate Execution Matrix” auditable rather than decorative.

Reframing the 3–6–9 module

Keep the modulo-9 routine as an experimental feature called a *Digital Root Classification*, not as a proved causal law. Store both the raw representation and a normalized variant. For example, use integer ticks rather than the formatted display price, so 123.45 and 12345 cents are not silently conflated. Then test whether roots or root transitions add value beyond price level, volatility, and trend. If the effect does not survive multiple instruments and held-out periods, downgrade it to visualization only.

Example: disciplined long setup

A valid bullish GSPS setup should require more than “root = 3.” Example logic: a confirmed daily swing low anchors the price engine; a projected Square-of-Nine-derived level lies within a predefined tolerance; price reclaims that level after a failed breakdown; trend state turns upward; relative volume exceeds its threshold; and the current bar falls inside a predeclared time window. Entry occurs only above the reclaim bar’s high; invalidation lies below the failed-breakdown low or a volatility-defined stop. The 3/6/9 feature can be reported as contextual metadata, but it should not overrule price, time, volume, and risk.

Research and validation plan

The essential question is not “does Gann work?” It is “which precisely defined GSPS features improve decision quality after costs, across assets and market regimes, compared with simpler alternatives?”

Phase 1: Define before testing

- Write a formal specification for every calculation: pivot rule, anchor choice, angle normalization, square transformation, time intervals, tolerances, entry trigger, stop, target, and exit expiry.
- Freeze definitions and parameter ranges before measuring results. Do not choose a cycle, rotation, or tolerance after observing favorable historical outcomes.
- Create a feature registry distinguishing: historically documented Gann concept; modern interpretation; GSPS-native convention; and experimental hypothesis.

Phase 2: Build comparable baselines

- Benchmark GSPS against simple alternatives: breakout plus ATR stop, moving-average trend with relative-volume confirmation, prior-swing support/resistance, and random time-matched controls.
- For each Gann level or time window, generate matched control levels/windows with the same volatility, trend, and distance-to-price distribution. This tests whether the Gann construction adds information rather than merely marking places price often visits.

Phase 3: Use robust evaluation

- Separate in-sample design from locked out-of-sample evaluation. Use walk-forward windows and do not tune on the final test period.
- Include spreads, commissions, realistic slippage, borrow constraints for shorts, liquidity filters, earnings/event rules, and execution delay.
- Report hit rate, average win/loss, expectancy in R, profit factor, maximum drawdown, Sharpe or Sortino ratio where appropriate, turnover, capacity, and performance by volatility/trend regime.
- Use bootstrap confidence intervals and multiple-testing control. If a 3/6/9 effect is tested across many timeframes, assets, and parameters, apparent winners may arise by chance.

Phase 4: Operational gate

- Promote a module from experimental to production only if it improves a predeclared out-of-sample metric over its baseline, remains stable across instruments and eras, and does not materially worsen tail risk.
- Keep the live system in decision-support or paper-trading mode until the research ledger shows durability. A visually convincing chart example is not validation.

Priority roadmap

Priority	Build item	Reason
P0	Repository-level inventory and traceability	Without the actual codebase and a feature register, implementation coverage cannot be measured precisely.
P1	Pivot/trend + OHLCV/volume confirmation	This is closest to Gann's practical, documented tape-reading work and supplies the context absent from root-only outputs.
P2	Risk engine and deterministic signal contract	Turns “entry/target/stop” labels into reproducible, risk-bounded decisions.
P3	Square of Nine and normalized angle modules	Implements the recognizable Gann framework while exposing formulas for independent testing.
P4	Time-cycle confluence and backtest harness	Tests whether timing concepts add value when combined with price structure.

Priority	Build item	Reason
P5	3/6/9 digital-root study	Maintain as research/visualization until it clears stringent out-of-sample tests.

Final judgment

Gann can credibly serve as GSPS’s philosophical foundation if “foundation” means a disciplined focus on the interaction of price, time, range, trend, and execution. It cannot responsibly mean that a single modulo-9 classification determines market direction or exact coordinates. The strongest version of GSPS is therefore a modern, falsifiable Gann research platform: it turns Gann’s ideas into transparent features, requires market confirmation, limits downside through formal risk logic, and retains only the components that demonstrate repeatable value.

Source notes

Primary historical material: W. D. Gann, *Truth of the Stock Tape* (1923), full text hosted by Internet Archive. The passages used in this report address tape reading, supply/demand, volume, price movement, time, resistance, and operating rules. [web:21]

Contemporaneous profile: Richard D. Wyckoff, *Ticker and Investment Digest*, December 1909, as reproduced in the retrieved source. Performance statements are reported as claims of that profile, not treated as audited evidence. [web:26]

Methodology context: modern explanatory sources commonly organize Gann practice around price, time, and pattern studies, but this report prioritizes the operational content in Gann’s own 1923 writing over later interpretations. [web:4][web:23]

Code-audit note: GitHub repository search returned no identifiable GSPS trading-system repository, and a broad code search returned no reliable GSPS-specific Gann implementation. This report’s coverage estimates are therefore constrained to the supplied protocol and should be updated against the actual repository. [github_mcp_direct:1][github_mcp_direct:2]